Linear Regression — Complete Derivation (Closed-Form & Gradient Descent)

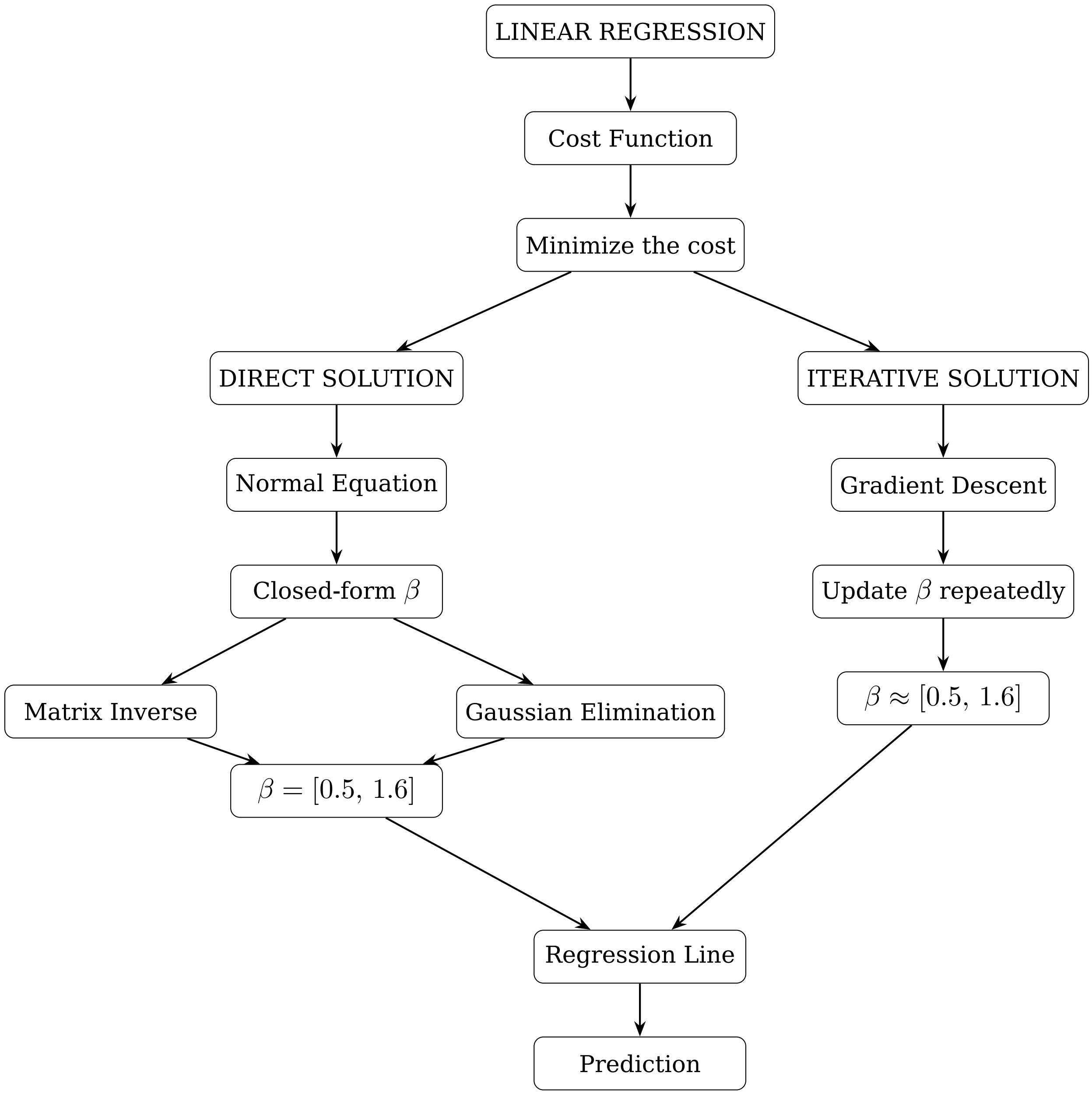
Contents

Dataset used throughout this document

$$(1, 2), \quad (2, 4), \quad (3, 5), \quad (4, 7)$$

Every method below — closed-form and gradient descent — is applied to this same dataset, so the final results can be directly compared.

Overview Flowchart



This diagram maps directly onto the document: **Part I** builds the cost function, **Part II** follows the left (direct) branch down to the exact $\beta = [0.5, 1.6]$ via either matrix inversion or Gaussian elimination, **Part IV** follows the right (iterative) branch down to an approximate $\beta \approx [0.5, 1.6]$ via repeated updates, and **Part III/V** cover where both branches reconverge: the regression line and the final prediction.

Part I — Setup: From Raw Data to the Cost Function

Step 1 — Dataset

Four data points, each (x_i, y_i):

$$(1, 2), \quad (2, 4), \quad (3, 5), \quad (4, 7)$$

$n = 4$ observations.

Terms: data point, sample size n.

Step 2 — Plot

Plotting the four points shows them rising roughly steadily as x increases — an approximately straight-line (linear) trend, which is why a linear model is a reasonable choice.

Terms: scatter plot, trend, linearity assumption.

Step 3 — Vectors

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 7 \end{bmatrix}$$

Terms: feature vector, target vector, column vector, transpose (^{T}).

Step 4 — Hypothesis

$$\hat{y} = \beta_0 + \beta_1 x$$

- $\hat{y}$ = predicted value
- β_0 = intercept
- β_1 = slope

Terms: hypothesis function, predicted value $\hat{y}$, intercept, slope.

Step 5 — Design Matrix

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

Terms: design matrix, bias/intercept column.

Step 6 — Parameter Vector

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

Terms: parameter vector, weights, coefficients.

Step 7 — Model in Matrix Form (Linear Combination)

$$\hat{y} = X\beta = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 \\ \beta_0 + 2\beta_1 \\ \beta_0 + 3\beta_1 \\ \beta_0 + 4\beta_1 \end{bmatrix}$$

Each row is a **linear combination** of the columns of X.

Terms: matrix-vector product, linear combination.

Step 8 — Linear Independence (full proof)

Terms introduced here: linear combination, trivial solution, linear independence/dependence, invertible matrix, singular matrix, determinant, rank.

Columns of X:

$$c_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \qquad c_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Definition. c_1, c_2 are linearly independent if the only solution to $k_1 c_1 + k_2 c_2 = \mathbf{0}$ is the trivial one, $k_1 = 0, k_2 = 0$.

Set up the equation:

$$k_1 \begin{bmatrix}1\\1\\1\\1\end{bmatrix} + k_2 \begin{bmatrix}1\\2\\3\\4\end{bmatrix} = \begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

giving four scalar equations:

$$k_1 + k_2 = 0 \ (1) \qquad k_1 + 2k_2 = 0 \ (2) \qquad k_1 + 3k_2 = 0 \ (3) \qquad k_1 + 4k_2 = 0 \ (4)$$

Subtract (1) from (2):

$$(k_1 + 2k_2) - (k_1 + k_2) = 0 \implies k_2 = 0$$

Substitute into (1):

$$k_1 + 0 = 0 \implies k_1 = 0$$

Check (3),(4): $0 + 3(0) = 0$ (OK), $0 + 4(0) = 0$ (OK).

Conclusion: only the trivial solution exists, so c_1, c_2 are linearly independent — this will guarantee $X^T X$ is invertible (confirmed numerically in Step 13 of Part II).

Terms: trivial solution, linear independence, invertibility.

Step 9 — Error and Cost Function

Error (residual) vector:

$$e = y - \hat{y} = y - X\beta$$

Cost function (mean squared error, matrix form — using $e^T e = \sum e_i^2$):

$$J(\beta) = \frac{1}{2n}\, e^T e = \frac{1}{2n}(y - X\beta)^T (y - X\beta)$$

This is a **quadratic function** of β — graphed against β_0, β_1 it forms a smooth bowl (paraboloid) with exactly one minimum.

Terms: error vector, cost function, quadratic form, convex function.

Part II — Method A: The Closed-Form Solution

Step 10 — Expand the Cost Function

Use $(A-B)^T(A-B) = A^TA - A^TB - B^TA + B^TB$ with $A = y,\ B = X\beta$:

$$J(\beta) = \frac{1}{2n}\Big(y^Ty - y^TX\beta - (X\beta)^Ty + (X\beta)^T(X\beta)\Big)$$

Since $(X\beta)^T = \beta^TX^T$, the middle two terms ($y^TX\beta$ and β^TX^Ty) are equal scalars and combine:

$$J(\beta) = \frac{1}{2n}\Big(y^Ty - 2\beta^TX^Ty + \beta^TX^TX\beta\Big)$$

Terms: transpose of a product, scalar equals its own transpose.

Step 11 — Differentiate the Cost Function

Using the matrix-calculus identities $\dfrac{\partial}{\partial\beta}(\beta^Ta) = a$ and $\dfrac{\partial}{\partial\beta}(\beta^TA\beta) = 2A\beta$ (for symmetric A):

$$\frac{\partial J}{\partial\beta} = \frac{1}{2n}\Big(-2X^Ty + 2X^TX\beta\Big)$$

Terms: matrix calculus, gradient with respect to a vector.

Step 12 — Set the Derivative to Zero (Normal Equation)

At the minimum, the gradient is zero:

$$\frac{1}{2n}\Big(-2X^Ty + 2X^TX\beta\Big) = 0 \implies -X^Ty + X^TX\beta = 0$$

$$\boxed{X^TX\,\beta = X^Ty} \quad \longleftarrow \ \textbf{Normal Equation}$$

Terms: normal equation, first-order condition, stationary point.

Step 13 — Solve for β (General Formula)

Left-multiply both sides by $(X^T X)^{-1}$ (valid because Step 8 proved the columns of X are linearly independent, so $X^T X$ is invertible):

$$\boxed{\beta = (X^T X)^{-1} X^T y}$$

Terms: closed-form solution, matrix inverse, identity matrix.

Step 14 — Compute $X^T X$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

- $(1,1)$: $1(1) + 1(1) + 1(1) + 1(1) = 4$
- $(1,2)$: $1(1) + 1(2) + 1(3) + 1(4) = 1 + 2 + 3 + 4 = 10$
- $(2,1)$: same as $(1,2)$ by symmetry $= 10$
- $(2,2)$: $1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$

$$\boxed{X^T X = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}}$$

Determinant check: $\det(X^T X) = (4)(30) - (10)(10) = 120 - 100 = 20 \neq 0$, confirming the inverse exists.

Step 15 — Compute $X^T y$

- Row 1: $1(2) + 1(4) + 1(5) + 1(7) = 2 + 4 + 5 + 7 = 18$
- Row 2: $1(2) + 2(4) + 3(5) + 4(7) = 2 + 8 + 15 + 28 = 53$

$$\boxed{X^T y = \begin{bmatrix} 18 \\ 53 \end{bmatrix}}$$

Step 16 — Compute $(X^T X)^{-1}$

For $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is $\dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. With $a = 4, b = 10, c = 10, d = 30$ and $\det = 20$:

$$(X^T X)^{-1} = \frac{1}{20} \begin{bmatrix} 30 & -10 \\ -10 & 4 \end{bmatrix} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix}$$

Terms: determinant, 2×2 matrix inverse formula.

Step 17 — Compute $\beta = (X^T X)^{-1} X^T y$

$$\beta = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 18 \\ 53 \end{bmatrix}$$

Row 1: $1.5(18) + (-0.5)(53) = 27 - 26.5 = 0.5$

Row 2: $-0.5(18) + 0.2(53) = -9 + 10.6 = 1.6$

$$\boxed{\beta_0 = 0.5} \qquad \boxed{\beta_1 = 1.6}$$

Step 18 — Alternative Route: Gaussian Elimination (sanity check)

The normal equation from Step 12, written as scalar equations:

$$4\beta_0 + 10\beta_1 = 18 \quad (i) \qquad 10\beta_0 + 30\beta_1 = 53 \quad (ii)$$

From (i): $\beta_0 = \dfrac{18 - 10\beta_1}{4} = 4.5 - 2.5\beta_1$

Substitute into (ii): $10(4.5 - 2.5\beta_1) + 30\beta_1 = 53 \implies 45 - 25\beta_1 + 30\beta_1 = 53 \implies 45 + 5\beta_1 = 53 \implies 5\beta_1 = 8$

$$\beta_1 = \frac{8}{5} = 1.6 \qquad \Rightarrow \qquad \beta_0 = 4.5 - 2.5(1.6) = 4.5 - 4 = 0.5$$

Matches Step 17 exactly — as it must, since both are solving the identical normal equation via different arithmetic routes (matrix inversion vs. elimination).

Terms: elimination, substitution, back-substitution.

Part III — Results Common to Both Methods

Step 19 — Regression Line

$$\boxed{\hat{y} = 0.5 + 1.6x}$$

Step 20 — Training Predictions

x	Calculation	$\hat{y}$
1	$0.5 + 1.6(1)$	2.1
2	$0.5 + 1.6(2)$	3.7
3	$0.5 + 1.6(3)$	5.3
4	$0.5 + 1.6(4)$	6.9

Step 21 — Residuals

x	y	$\hat{y}$	$e = y - \hat{y}$
1	2	2.1	-0.1
2	4	3.7	$+0.3$
3	5	5.3	-0.3
4	7	6.9	$+0.1$

(Residuals sum to zero — guaranteed for least-squares fits with an intercept.)

Step 22 — Minimum Cost

$$J = \frac{1}{8}\left[(-0.1)^2 + (0.3)^2 + (-0.3)^2 + (0.1)^2\right] = \frac{0.20}{8} = \boxed{0.025}$$

This is the **global minimum** — the closed-form solution guarantees this exactly.

Step 23 — New Input

$$x = 5 \quad \text{(beyond the training range 1–4, so this is extrapolation)}$$

Step 24 — Final Prediction (Closed-Form)

$$\hat{y} = 0.5 + 1.6(5) = 0.5 + 8.0 = \boxed{8.5}$$

Part IV — Method B: Gradient Descent (Iterative Alternative)

Step 25 — Why an Alternative Method?

The closed-form solution above is exact but relies on inverting $X^T X$ — expensive or unstable for very large numbers of features, and impossible for non-linear models (logistic regression, neural networks). **Gradient descent** solves the same minimization problem iteratively instead, and generalizes to those harder cases.

Terms: iterative solution, scalability.

Step 26 — Gradient Formulas (from calculus)

Differentiating $J(\beta_0, \beta_1) = \frac{1}{2n} \sum (y_i - \beta_0 - \beta_1 x_i)^2$ with respect to each parameter separately (chain rule):

$$\boxed{\frac{\partial J}{\partial \beta_0} = -\frac{1}{n} \sum (y_i - \hat{y}_i)} \qquad \boxed{\frac{\partial J}{\partial \beta_1} = -\frac{1}{n} \sum (y_i - \hat{y}_i)\, x_i}$$

Terms: gradient, partial derivative, chain rule.

Step 27 — Update Rule and Initialization

$$\beta_0 := \beta_0 - \alpha \frac{\partial J}{\partial \beta_0}, \qquad \beta_1 := \beta_1 - \alpha \frac{\partial J}{\partial \beta_1}$$

Learning rate: $\alpha = 0.1$. Initialization: $\beta_0 = 0,\ \beta_1 = 0$.

Terms: update rule, learning rate, initialization.

Step 28 — Iteration 1 (fully worked)

Predictions ($\beta_0 = \beta_1 = 0$): $\hat{y} = [0, 0, 0, 0]$

Errors: $e = [2, 4, 5, 7]$

Cost: $J = \frac{1}{8}(4 + 16 + 25 + 49) = \frac{94}{8} = 11.75$

Gradient for β_0: $-\frac{1}{4}(2 + 4 + 5 + 7) = -\frac{18}{4} = -4.5$

Gradient for β_1: $-\frac{1}{4}(2(1) + 4(2) + 5(3) + 7(4)) = -\frac{53}{4} = -13.25$

Update:

$$\beta_0 := 0 - 0.1(-4.5) = 0.45 \qquad \beta_1 := 0 - 0.1(-13.25) = 1.325$$

Step 29 — Iteration 2 (fully worked)

Predictions ($\beta_0 = 0.45, \beta_1 = 1.325$): $\hat{y}_1 = 1.775,\ \hat{y}_2 = 3.10,\ \hat{y}_3 = 4.425,\ \hat{y}_4 = 5.75$

Errors: $e_1 = 0.225,\ e_2 = 0.90,\ e_3 = 0.575,\ e_4 = 1.25$

Cost: $J = \frac{1}{8}(0.050625 + 0.81 + 0.330625 + 1.5625) = \frac{2.75375}{8} \approx 0.3442$

Gradient for β_0: $-\frac{1}{4}(0.225 + 0.90 + 0.575 + 1.25) = -\frac{2.95}{4} = -0.7375$

Gradient for β_1: $-\frac{1}{4}(0.225(1) + 0.90(2) + 0.575(3) + 1.25(4)) = -\frac{8.75}{4} = -2.1875$

Update:

$$\beta_0 := 0.45 + 0.07375 = 0.52375 \qquad \beta_1 := 1.325 + 0.21875 = 1.54375$$

Step 30 — Iteration 3 (fully worked)

Predictions ($\beta_0 = 0.52375, \beta_1 = 1.54375$): $\hat{y}_1 = 2.0675,\ \hat{y}_2 = 3.61125,\ \hat{y}_3 = 5.155,\ \hat{y}_4 = 6.69875$

Errors: $e_1 = -0.0675,\ e_2 = 0.38875,\ e_3 = -0.155,\ e_4 = 0.30125$

Cost: $J \approx \frac{0.27045}{8} \approx 0.0338$

Gradient for β_0: ≈ -0.1169

Gradient for β_1: ≈ -0.3625

Update:

$$\beta_0 \approx 0.52375 + 0.01169 \approx 0.5354 \qquad \beta_1 \approx 1.54375 + 0.03625 \approx 1.58$$

Step 31 — Convergence Over More Iterations

Iteration	β_0 (start)	β_1 (start)	Cost J	β_0 (after)	β_1 (after)
1	0.0000	0.0000	11.75000	0.4500	1.3250
2	0.4500	1.3250	0.34422	0.5238	1.5438
3	0.5238	1.5438	0.03381	0.5354	1.5800
5	0.5369	1.5861	0.02512	0.5367	1.5873
10	0.5346	1.5882	0.02510	0.5341	1.5884
20	0.5297	1.5899	0.02507	0.5293	1.5900

Iteration	β_0 (start)	β_1 (start)	Cost J	β_0 (after)	β_1 (after)
30	0.5256	1.5913	0.02505	0.5252	1.5914

Both parameters slowly approach $(0.5,\ 1.6)$ — the exact closed-form answer from Step 17/18 — and the cost creeps toward the true minimum $J = 0.025$ from Step 22.

Terms: convergence, plateauing, fixed learning rate.

Step 32 — Final Prediction (Gradient Descent, after 30 iterations)

$$\hat{y} = 0.5256 + 1.5913(5) = 0.5256 + 7.9565 \approx \boxed{8.482}$$

Compare to the exact closed-form prediction: $\hat{y} = 8.5$ (Step 24). Gradient descent is close but not exact after only 30 steps; more iterations (or a decaying learning rate) would close the remaining gap.

Part V — Side-by-Side Comparison

	Closed-Form (Steps 10–18)	Gradient Descent (Steps 25–32)
Formula	$\beta = (X^T X)^{-1} X^T y$	$\beta := \beta - \alpha \nabla J$, repeated
Result for β_0, β_1	Exact: $0.5,\ 1.6$	Approximate after 30 iters: $0.5256,\ 1.5913$
Cost achieved	Exact minimum: 0.025	Approaching: 0.02505
Prediction at $x = 5$	Exact: 8.5	Approximate: ≈ 8.482
Needs learning rate?	No	Yes
Needs initialization?	No	Yes
Cost to compute	$O(p^3)$ matrix inversion	$O(np)$ per iteration, many iterations
Scales to large feature counts?	Poorly	Well
Works for non-linear models?	No	Yes
Requires invertibility of $X^T X$?	Yes (proved in Step 8)	No

Bottom line: for this small, well-conditioned dataset, the closed-form solution is exact, faster, and requires no tuning. Gradient descent converges toward the same answer but needs many iterations (and a tuned learning rate) to get there — its real value shows up on large-scale or non-linear problems where the closed-form route is impractical or doesn't exist.